Module 4: BIG DATA ANALYTICS & DISTRIBUTED COMPUTING

Course Code: 22DS102006 - Apache Spark, PySpark, Hadoop & Streaming Systems

1. Distributed Computing Fundamentals

The Hadoop Distributed File System (HDFS) and MapReduce paradigm. Master-worker topologies, replication strategies, and fault tolerance across distributed compute commodity clusters.

2. In-Memory Distributed Analytics with Apache Spark

Resilient Distributed Datasets (RDDs), DAG execution planners, lazy evaluation, transformations vs actions, and Spark SQL DataFrames. Hands-on PySpark scripting for processing terabyte datasets.

3. Real-Time Event Streaming

Apache Kafka architecture: topics, partitions, broker clusters, producer/consumer offsets. Spark Structured Streaming for low-latency windowed aggregations on streaming sensor logs.